

$$f(x) = a^x = a^2 + (x-2)a \log a + \frac{(x-2)^2}{2!} a^2 (\log a)^2 + \frac{(x-2)^3}{3!} a^3 (\log a)^3 + \dots + \frac{(x-2)^n}{n!} a^n (\log a)^n$$

Hand written Sheet last ques.

If $y^{1/m} + y^{-1/m} = 2x \sqrt{1-x^2}$ — (1)

prove that $(x^2-1)^m y^{m+2} + (2m+1)x^m y^{m+1} + (m^2-m^2) y^m = 0$

Let $y^{1/m} = z$ so $y^{-1/m} = \frac{1}{z}$ put in (1)

$$\Rightarrow z + \frac{1}{z} = 2x$$

$$\Rightarrow \frac{z^2+1}{z} = 2x$$

$$\Rightarrow z^2 - 2xz + 1 = 0 \quad (\text{quadratic eq in } z)$$

$$\Rightarrow z = \frac{2x \pm \sqrt{4x^2-4}}{2} = x \pm \sqrt{x^2-1}$$

$$z = x \pm \sqrt{x^2-1}$$

$$\text{Let } z = x + \sqrt{x^2-1}$$

$$\Rightarrow y^{1/m} = x + \sqrt{x^2-1}$$

$$\Rightarrow (y^{1/m})^m = (x + \sqrt{x^2-1})^m$$

$$y = (x + \sqrt{x^2-1})^m \quad \text{--- (2)}$$

Diff w.r.t x, we get.

$$y_1 = m(x + \sqrt{x^2-1})^{m-1} \left[x + \frac{x}{\sqrt{x^2-1}} \right]$$

$$y_1 = m(x + \sqrt{x^2-1})^{m-1} \left(\frac{\sqrt{x^2-1} + x}{\sqrt{x^2-1}} \right)$$

$$(\sqrt{x^2-1}) y_1 = m(x + \sqrt{x^2-1})^m$$

sq.ing, we get

$$(x^2-1) y_1^2 = m^2 (x + \sqrt{x^2-1})^{2m}$$

$$(x^2-1) y_1^2 \Rightarrow m^2 y_1^2$$

$$(x^2-1) y_1^2 - m^2 y_1^2 = 0 \quad \text{--- (3)}$$

Diff (3), we get

$$(x^2-1) y_2 + x y_1 - m^2 y = 0 \quad \text{--- (4)}$$

$$\Rightarrow y_{m+2}(x^2-1) + 2mx y_{m+1} + m(m-1) y_m + y_{m+1} x + m y_m - m^2 y_m = 0$$

$$\Rightarrow (x^2-1) y_{m+2} + (2m+1)x y_{m+1} + (m^2-m^2) y_m = 0$$

① Ques Taylor's. then

Expand $\ln \cos x$ in powers of $(x - \pi/3)$

(point abt = $\pi/3$)

$$f(x) = f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \frac{h^3}{3!} f'''(a) + \frac{h^4}{4!} f^{(4)}(a) + \dots$$

$$+ \frac{h^n}{n!} f^{(n)}(a+0h) \quad \rightarrow (1)$$

[Here, $h = x - \pi/3$, $a = \pi/3$]

$$f(x) = \ln \cos x = f\left(\frac{\pi}{3}\right) + \left(x - \frac{\pi}{3}\right) f'\left(\frac{\pi}{3}\right) + \frac{\left(x - \frac{\pi}{3}\right)^2}{2!} f''\left(\frac{\pi}{3}\right) + \frac{\left(x - \frac{\pi}{3}\right)^3}{3!} f'''\left(\frac{\pi}{3}\right) +$$

$$\frac{\left(x - \frac{\pi}{3}\right)^4}{4!} f^{(4)}\left(\frac{\pi}{3}\right) + \dots + \frac{\left(x - \frac{\pi}{3}\right)^n}{n!} f^{(n)}\left(\frac{\pi}{3}\right) + \frac{\left(x - \frac{\pi}{3}\right)^{n+1}}{(n+1)!} f^{(n+1)}\left(\frac{\pi}{3}\right) \quad \text{--- (2)}$$

$$f(x) = \ln \cos x \Rightarrow \log(1/2)$$

$$f'(x) = -\frac{\sin x}{\cos x} = -\tan x \quad \text{at } f\left(\frac{\pi}{3}\right) \Rightarrow \text{--- (3)}$$

$$f''(x) = -\sec^2 x \quad \text{at } f\left(\frac{\pi}{3}\right) \Rightarrow \text{--- (4)}$$

$$f'''(x) = -2\sec x \tan x \quad \text{at } f\left(\frac{\pi}{3}\right) = \text{--- (5)}$$

$$f^{(4)}(x) = -2\sec^2 x \sec^2 x - 2\tan^2 x \cdot 2\sec^2 x$$

$$= -2\sec^4 x - 4\sec^2 x \tan^2 x$$

$$f^{(4)}(x) \text{ at } \left(\frac{\pi}{3}\right) \Rightarrow \text{--- (6)}$$

$$f(x) = \ln \cos x = \ln \left(\frac{1}{2} \right) + \left(\frac{x-\pi}{3} \right) (-\sqrt{3}) + \frac{1}{2!} \left(\frac{x-\pi}{3} \right)^2 (-4) + \frac{1}{3!} \left(\frac{x-\pi}{3} \right)^3 (8\sqrt{3}) + \dots$$

$$\frac{1}{4!} \left(\frac{x-\pi}{3} \right)^4 (-80) + \dots$$

$$f(x) = \ln \cos x = -\ln 2 + \left(\frac{x-\pi}{3} \right) (-\sqrt{3}) - \frac{2}{3} \left(\frac{x-\pi}{3} \right)^2 - \frac{4\sqrt{3}}{3} \left(\frac{x-\pi}{3} \right)^3 - \frac{10}{3} \left(\frac{x-\pi}{3} \right)^4 + \dots$$

② Expand $\tan(x + \pi/4)$ as far as the term x^4 & hence evaluate $\tan 46.5^\circ$.

Ans: The Taylor's formula given by:-

$$f(x) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \frac{h^3}{3!} f'''(a) + \frac{h^4}{4!} f^{(4)}(a) + \dots$$

$$h^n f^n(a + \theta h) \quad \text{--- (1)}$$

Here $a = \pi/4$, $h = x$

$$f(x) = \tan x$$

$$f(x) = \tan x = f\left(\frac{\pi}{4}\right) + x f'\left(\frac{\pi}{4}\right) + \frac{x^2}{2!} f''\left(\frac{\pi}{4}\right) + \frac{x^3}{3!} f'''(\pi/4) + \frac{x^4}{4!} f^{(4)}(\pi/4) + \dots$$

$$f(x) = \tan x \quad \text{at } \left(\frac{\pi}{4}\right) = 1$$

$$f'(x) = \sec^2 x \quad \text{at } \left(\frac{\pi}{4}\right) = 2$$

$$f''(x) = 2 \sec^2 x \tan x \quad \text{at } \left(\frac{\pi}{4}\right) = 4$$

$$f'''(x) = 2 \sec^4 x + 4 \sec^2 x \tan^2 x \quad \text{at } \left(\frac{\pi}{4}\right) = 16$$

$$f^{(4)}(x) = 8 \quad \text{at } \left(\frac{\pi}{4}\right) = 80$$

$$f(x) = \tan x = 1 + \frac{1}{1}x + \frac{2}{2}x^2 + \frac{4}{3}x^3 + \frac{10}{3}x^4 + \dots \quad (3)$$

$$180^\circ = \pi \text{ rad.}$$

$$1^\circ \Rightarrow \frac{\pi}{180} \text{ rad.}$$

$$180$$

$$1.5^\circ \Rightarrow 0.02618 \text{ rad.}$$

$$\Rightarrow \tan \left(x + \frac{\pi}{4} \right)$$

$$\Rightarrow \tan (45 + 1.5)$$

$$\tan \left(\frac{\pi}{4} + 0.02618 \right)$$

putting, $x = 0.02618 \text{ in (3)}$, we get

$$\tan(46.5^\circ) = 1 + 2(0.02618) + 2(0.02618)^2 + \frac{8}{3}(0.02618)^3 + \frac{10}{3}(0.02618)^4 + \dots$$

$$\tan(46.5^\circ) \approx 1.05378.$$

Practice Set 1

①. $f(x) = x^2$ $[a, b] = [-1, 1]$

$f(x) = x^2$ is a polynomial f^n . All poly. f^n are continuous and differentiable everywhere in closed interval $[-1, 1]$ and open interval $(-1, 1)$ respectively.

$f(1) = (1)^2 = 1$ $[f(a) = f(b)]$
 $f(1) = (1)^2 = 1$ $[1 = 1]$

$f'(x) = 2x = 0$
 $f'(x) = 2x = 0$
 $= [x = 0]$ Hence Rolle's thm. verified.

②. $f(x) = x + 1/x$ $x \in [1/2, 3]$

$f'(x) = f(b) - f(a)$ $f(a) = f(1/2) = 1 + 2 + 0 = 5/2$
 $f'(x) = \frac{10 - 5}{3 - 1/2} = \frac{5}{5/2} = 2$ $f(b) = f(3) = 3 + 1/3 = 10/3$

$f'(x) = \frac{5}{6} = 1/3$

$1 - 1 = 0$
 $c^2 = 3$

$1 - 1 = 0$
 $c^2 = 3/2$

$c = \pm \sqrt{3/2}$

③. $f(x) = e^x$, $g(x) = e^{-x}$

(MVT gives $-\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$)

$f'(x) = e^x$, $g'(x) = -e^{-x}$

$\Rightarrow \frac{e^b - e^a}{e^{-b} - e^{-a}} = \frac{e^c}{-e^{-c}}$

$\Rightarrow -e^{a+b} = -e^{2c}$

$a + b = 2c$
 $c = \frac{a+b}{2}$

④. (i) $y = f(x) = \sin 2x \cos 3x$ $\therefore \sin a \cos b = \frac{1}{2} [\sin(a+b) + \sin(a-b)]$

$y = f(x) = \frac{1}{2} (\sin 5x + (-) \sin x)$

$y = f(x) = \frac{1}{2} (\sin 5x - \sin x)$

$y'' = \frac{5^4 \sin(5x + \pi/2)}{2} - \frac{1^4 \sin(x + \pi/2)}{2}$

(ii) $y = e^x \sin^2 x$ $\therefore \sin^2 x = \frac{1 - \cos 2x}{2}$

$y = e^x (1 - \cos 2x)$

$y' = e^x - e^x \cos 2x$

$$e^{bx} (bx+c)$$

$$y = \frac{e^x}{2} - \frac{e^x \cos(2x)}{2}$$

$$e^{ax} \cos(bx) = y^n e^{ax} \cos(2x + n\phi)$$

$$\left[y_0 \sqrt{a^2 + b^2} \right]$$

$$y_n = \frac{e^x}{2} - \frac{1}{2} \left[y^n e^x \cos(2x + n\phi) \right]$$

$$y_n = \frac{e^x}{2} - \frac{1}{2} \left[(1+y)^n e^x \cos(2x + n \tan^{-1}(2)) \right]$$

$$y_n = \frac{e^x}{2} - \frac{e^x 5^{n/2} \cos(2x + n \tan^{-1}(2))}{2}$$

$$(iii) y = \ln(ax+u^2)$$

$$\therefore \log(ax+b) = \frac{a^n (-1)^{n-1} (n-1)!}{(ax+b)^n}$$

$$y = \ln(x(a+x))$$

$$y_n = \frac{(-1)^{n-1} (n-1)!}{(x)^n} + \frac{(-1)^{n-1} (n-1)!}{(x+a)^n}$$

$$y_n = \frac{(-1)^{n-1} (n-1)!}{x^n} \left[\frac{1}{x^n} + \frac{1}{(x+a)^n} \right]$$

$$(5) y(x) = a \cos(\log x) + b \sin(\log x)$$

$$y'(x) = -a \sin(\log x) + b \cos(\log x)$$

$$y''(x) = -\frac{a \cos(\log x)}{x^2} - \frac{b \sin(\log x)}{x^2}$$

$$y''(x) = -\left[\frac{y(x)}{x^2} \right]$$

$$y_1 \cdot x^2 + y = 0$$

$$(5) y = a \cos(\log x) + b \sin(\log x)$$

$$y_1 = -\frac{a \sin(\log x)}{x} + \frac{b \cos(\log x)}{x}$$

$$y_1 \cdot x = -a \cos \log x - \frac{b \sin(\log x)}{x}$$

$$x(y_2 x + y_1) = -y$$

$$y_2 x^2 + y_1 x + y = 0$$

$$y_0 y \cdot x^2 + y_1 y_{n+1} (2x) + y_2 y_n (2) + y (x) + y_n (n) + y_n = 0$$

$$x^2 y_{n+2} + n \cdot 2x y_{n+1} + n(n-1) y_n (2) + x y_{n+1} + n y_n + y_n = 0$$

$$(8) y = e^{ax \sin^{-1} x}$$

$$y' = e^{ax \sin^{-1} x} \cdot ax \cdot \frac{1}{\sqrt{1-x^2}} \Rightarrow y_1 = \frac{axy}{\sqrt{1-x^2}}$$

$$\rightarrow y_1(\sqrt{1-x^2}) = my$$

$$\rightarrow y_1^2(1-x^2) = m^2y^2$$

$$\rightarrow -2xy_1^2 + (1-x^2)2y_1y_2 = 2m^2yy_1$$

$$\rightarrow \frac{2y_1}{(1-x^2)y_2 - xy_1 - m^2y = 0}$$

$$y_{n+2}(1-x^2) + y_{n+1}(n)(-2x) + y_n \frac{n(n-1)}{2}(-2x) + (ny_{n+1}) - ny_n$$

$$-m^2y_n = 0$$

$$\left[(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0 \right]$$

$$(y)_0 = e^0 = 1$$

$$(y)_1 = m$$

$$(y)_2 = m^2$$

$$(y_{n+2})_0 = (n^2+m^2)(y_n)_0$$

$$n=1, (y_3)_0 = (1+m^2)m$$

$$n=2, (y_4)_0 = (2^2+m^2)m^2$$

$$n=3, (y_5)_0 = (3^2+m^2)(1+m^2)m$$

$$n=4, (y_6)_0 = (4^2+m^2)(2^2+m^2)(m^2)$$

$$(y)_n = \int m^2(2^2+m^2)(4^2+m^2) \dots [(n-2)^2+m^2], \text{ n is even.}$$

$$m^2(1^2+m^2)(3^2+m^2) \dots [(n-1)^2+m^2], \text{ n is odd.}$$

Partial Differentiable :-

functions of several variables.

$$z = f(x, y)$$

↓
dependent variable 2 independent variables

1st order derivative :- $\frac{\partial z}{\partial x}$ or $\frac{dz}{dx}$ or f_x or f_x

$$\frac{\partial^2 z}{\partial y \partial x} \text{ or } \frac{\partial^2 z}{\partial x \partial y} \text{ or } f_{xy} \text{ or } f_{yx}$$

2nd order derivative :- $\frac{\partial^2 z}{\partial x^2} = f_{xx}$ or f_{xx}

$$\frac{\partial^2 z}{\partial y \partial x} = f_{yx} = f_{xy}$$

$$\frac{\partial^2 z}{\partial y^2} = f_{yy}$$

$$\frac{\partial^2 z}{\partial x \partial y} = f_{xy} = f_{yx}$$

If $z = f(x, y)$ possesses continuous 2nd order partial derivatives then $\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial^2 z}{\partial x \partial y}$

3rd order derivative :- $\frac{\partial^3 z}{\partial x^3} = f_{xxx}$

$$\frac{\partial^3 z}{\partial y \partial x^2} = f_{xyx} = f_{xxy}$$

$$\frac{\partial^3 z}{\partial x \partial y^2} = f_{xyy} = f_{yyx}$$

$$\frac{\partial^3 z}{\partial y \partial x \partial y} = f_{xyy} = f_{yyx}$$

$$\frac{\partial^2 z}{\partial x^2} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x, y) - f(x, y)}{\delta x}$$

$$\frac{\partial^2 z}{\partial y^2} = \lim_{\delta y \rightarrow 0} \frac{f(x, y + \delta y) - f(x, y)}{\delta y}$$

$$\textcircled{1} z = x^2y + e^{xy} + y^3$$

Q1 $u = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right)$ find the value of $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}$

Diff ① w.r.t x , we get

$$\frac{\partial u}{\partial x} = \frac{1}{\sqrt{1-\frac{x^2}{y^2}}} \left(\frac{1}{y}\right) + \left(\frac{-y}{x^2}\right) (x^2)$$

$$\frac{\partial u}{\partial x} = \frac{1}{\sqrt{y^2-x^2}} - \frac{y}{x^2+y^2}$$

$$\left[\frac{x \left(\frac{\partial u}{\partial x} \right) }{\sqrt{y^2-x^2}} - \frac{yx}{x^2+y^2} \right] \rightarrow \text{②}$$

Diff ① w.r.t y , we get

$$\frac{\partial u}{\partial y} = -\frac{xy}{x^2+y^2} + \frac{(y^2+x^2)x}{y^2}$$

$$\left[\frac{x \left(\frac{\partial u}{\partial y} \right) }{\sqrt{y^2-x^2}} + \frac{xy}{x^2+y^2} \right] \rightarrow \text{③}$$

Add eq ② and ③ we get

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{x}{\sqrt{y^2-x^2}} - \frac{yx}{x^2+y^2} - \frac{yx}{x^2+y^2} + \frac{xy}{\sqrt{y^2-x^2}}$$

$$\text{Ans} = 0$$

Q1 If $u = \log(\tan x + \tan y + \tan z)$, show that

$$\sin 2x \left(\frac{\partial u}{\partial x} \right) + \sin 2y \left(\frac{\partial u}{\partial y} \right) + \sin 2z \left(\frac{\partial u}{\partial z} \right) = 2$$

Diff ① w.r.t x , we get

$$\frac{\partial u}{\partial x} = \frac{\sec^2 x}{\tan x + \tan y + \tan z}$$

$$\frac{\partial u}{\partial y} = \frac{\sec^2 y}{\tan x + \tan y + \tan z}$$

$$\frac{\partial u}{\partial z} = \frac{\sec^2 z}{\tan x + \tan y + \tan z}$$

$$\text{Now, } \sin 2x \left(\frac{\partial u}{\partial x} \right) + \sin 2y \left(\frac{\partial u}{\partial y} \right) + \sin 2z \left(\frac{\partial u}{\partial z} \right) = 2$$

$$= \frac{2 \sin x \cos x}{\cos^2 x} + \frac{2 \sin y \cos y}{\cos^2 y} + \frac{2 \sin z \cos z}{\cos^2 z}$$

$$= \frac{2 (\tan x + \tan y + \tan z)}{\tan x + \tan y + \tan z}$$

$$= 2 \rightarrow \text{Ans}$$

Q if $u = \log(x^3 + y^3 + z^3 - 3xyz)$

show :- (i) $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$

(ii) $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = -\frac{9}{(x+y+z)^2}$

(iii) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} + 2\frac{\partial^2 u}{\partial x \partial y} + 2\frac{\partial^2 u}{\partial x \partial z} + 2\frac{\partial^2 u}{\partial y \partial z} = -\frac{9}{(x+y+z)^3}$

Q diff (1) w.r.t x, y, z resp. we get.

$\frac{\partial u}{\partial x} = \frac{3x^2 - 3yz}{x^3 + y^3 + z^3 - 3xyz} \rightarrow (2)$

$\frac{\partial u}{\partial y} = \frac{3y^2 - 3xz}{x^3 + y^3 + z^3 - 3xyz} \rightarrow (3)$

$\frac{\partial u}{\partial z} = \frac{3z^2 - 3xy}{x^3 + y^3 + z^3 - 3xyz} \rightarrow (4)$

Now add (2) (3) (4) we get

$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3x^2 + 3y^2 + 3z^2 - 3yz - 3xz - 3xy}{x^3 + y^3 + z^3 - 3xyz}$

$= \frac{3(x^2 + y^2 + z^2 - yz - xz - xy)}{x^3 + y^3 + z^3 - 3xyz}$

$= \frac{3(x^2 + y^2 + z^2 - yz - xz - xy)}{x^3 + y^3 + z^3 - 3xyz}$

Ques Taylor's thm.

(1) expand sine in power of $(x - \pi/2)$, with Lagrange's form of remainder.

$f(x) = f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \frac{h^3}{3!}f'''(a) + \dots + \frac{h^n}{n!}f^{(n)}(a+h\theta)$, $0 < \theta < 1$

$h = x - \frac{\pi}{2}$, $a = \frac{\pi}{2}$

$f(x) = f\left(\frac{\pi}{2}\right) + \left(x - \frac{\pi}{2}\right)f'\left(\frac{\pi}{2}\right) + \frac{\left(x - \frac{\pi}{2}\right)^2}{2!}f''\left(\frac{\pi}{2}\right) + \frac{\left(x - \frac{\pi}{2}\right)^3}{3!}f'''(\frac{\pi}{2}) + \dots + \frac{\left(x - \frac{\pi}{2}\right)^n}{n!}f^{(n)}\left(\frac{\pi}{2} + \theta\left(x - \frac{\pi}{2}\right)\right)$

$f(x) = \sin x$
 $f^{(n)}(x) = \sin(x + n\pi/2)$

$f\left(\frac{\pi}{2} + \left(x - \frac{\pi}{2}\right)\theta\right) = \sin\left[\frac{\pi}{2} + \left(x - \frac{\pi}{2}\right)\theta + n\pi/2\right] = \sin\left(n\pi/2 + \left(x - \frac{\pi}{2}\right)\theta\right)$

$f\left(\frac{\pi}{2}\right) = 1, \sin(90^\circ)$
 $f'\left(\frac{\pi}{2}\right) = 0, \cos(90^\circ)$
 $f''\left(\frac{\pi}{2}\right) = -1, -\sin(90^\circ)$
 $f'''(\frac{\pi}{2}) = 0, -\cos(90^\circ)$

Ans: $f(x) = 1 - \frac{1}{2}\left(x - \frac{\pi}{2}\right)^2 + \frac{1}{24}\left(x - \frac{\pi}{2}\right)^4 - \dots - \frac{\left(x - \frac{\pi}{2}\right)^n}{n!} \sin\left(n\pi/2 + \left(x - \frac{\pi}{2}\right)\theta\right)$, $0 < \theta < 1$

① Find the approx. value of $\sqrt{10}$ correct upto four decimal places by taking the first four terms of Taylor's expansion.

$$f(x) = f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \frac{h^3}{3!} f'''(a) + \dots$$

$$\sqrt{10} \rightarrow \sqrt{9+1} \quad \left[\begin{array}{l} a=9 \\ h=1 \\ f(x) = \sqrt{x} \end{array} \right]$$

$$f(9) = 3$$

$$f'(9) = \frac{1}{2\sqrt{9}} = \frac{1}{18}$$

$$f''(9) = \frac{-1}{4} \frac{1}{(9)^{3/2}} =$$

$$f'''(9) = \frac{3}{8} \frac{1}{(9)^{5/2}}$$

$$f(x) = 3 + \frac{1}{18} + \frac{1^2}{2}$$

② Expand e^x

$$f(x) = f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \frac{h^3}{3!} f'''(a) + \dots + \frac{h^n}{n!} f^{(n)}(a) + \dots$$

Taking $h=x, a=0$; (Using Maclaurin's thm)

$$f(x) = f(0) + hf'(0) + \frac{h^2}{2!} f''(0) + \frac{h^3}{3!} f'''(0) + \dots + \frac{h^n}{n!} f^{(n)}(0) + \dots$$

$$\text{Ans} \Rightarrow e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

③ Expand $\tan^{-1}(x+h)$ ($a=x, h=h$), (4 terms)

$$[\tan^{-1}(1.0003)], \text{ ans} = 0.78540]$$

$$a=1, h=0.0003$$